Activity 7 — Multi-Product Shopping Budget Constraint

1. Problem Statement

Develop a Python program to calculate the total cost of multiple products and check whether the shopping expenses are within the customer's budget.

2. Objectives

- To calculate the total shopping cost.
- To compare the total cost with the available budget.
- To identify whether the customer is within budget.
- To understand Python variables, arithmetic operators and conditional statements.

3. Assumptions

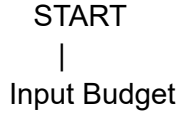
For this example, assume:

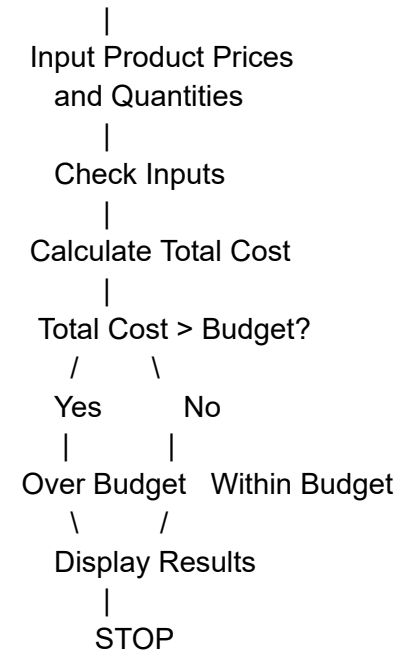
- The customer has a budget of ₹2,000.
- The customer purchases three types of products.
- The customer enters each product's price and quantity.
- Prices must be positive and quantities must be greater than zero.

4. Algorithm

1. Start.
2. Enter the available shopping budget.
3. Enter the price and quantity of three products.
4. Check whether the inputs are valid.
5. Calculate the cost of each product.
6. Calculate the total shopping cost.
7. Compare the total cost with the budget.
8. Display the total cost and budget status.
9. Stop.

5. Flowchart





6. Pseudocode

```
START
INPUT budget
INPUT price1, quantity1
INPUT price2, quantity2
INPUT price3, quantity3

IF any input is invalid
   PRINT "Invalid input"
ELSE
   cost1 = price1 * quantity1
   cost2 = price2 * quantity2
   cost3 = price3 * quantity3

   total = cost1 + cost2 + cost3

   PRINT total

   IF total > budget
      PRINT "Over Budget"
   ELSE
      PRINT "Within Budget"
   ENDIF
ENDIF

STOP
```

7. Python Source Code

File name: `activity7.py`

```
budget = float(input("Enter shopping budget: "))

price1 = float(input("Enter price of product 1: "))
quantity1 = int(input("Enter quantity of product 1: "))

price2 = float(input("Enter price of product 2: "))
quantity2 = int(input("Enter quantity of product 2: "))

price3 = float(input("Enter price of product 3: "))
quantity3 = int(input("Enter quantity of product 3: "))

if (budget < 0 or price1 <= 0 or price2 <= 0 or price3 <= 0
      or quantity1 <= 0 or quantity2 <= 0 or quantity3 <= 0):
   print("Invalid input")

else:
   cost1 = price1 * quantity1
   cost2 = price2 * quantity2
   cost3 = price3 * quantity3

   total = cost1 + cost2 + cost3

   print("Total Shopping Cost: Rs.", total)

   if total > budget:
      print("Status: Over Budget")
      print("Extra Amount Needed: Rs.", total - budget)
   else:
      print("Status: Within Budget")
      print("Remaining Budget: Rs.", budget - total)
```

8. Test Cases

Budget	Product Costs	Expected Result
₹2,000	₹500 + ₹300 + ₹200	₹1,000 total; ₹1,000 remaining
₹1,000	₹500 + ₹400 + ₹300	₹1,200 total; ₹200 over budget
₹2,000	₹200 + ₹300 + ₹500	₹1,000 total; ₹1,000 remaining
₹2,000	Negative product price	Invalid input

For these examples, each product quantity is 1.

9. Output Screenshot and Conclusion

Sample Output:

Enter shopping budget: 2000
Enter price of product 1: 500
Enter quantity of product 1: 1
Enter price of product 2: 300
Enter quantity of product 2: 1
Enter price of product 3: 200
Enter quantity of product 3: 1
Total Shopping Cost: Rs. 1000.0
Status: Within Budget
Remaining Budget: Rs. 1000.0

Output Screenshot: Run `activity7.py` in Python IDLE or VS Code. Capture the inputs and output, and save the screenshot as `activity7_output.png`.

Conclusion: This program calculates the total cost of multiple products and checks whether shopping expenses fit within the available budget.

INPUT:

```
budget = float(input("Enter shopping budget: "))

price1 = float(input("Enter price of product 1: "))
quantity1 = int(input("Enter quantity of product 1: "))

price2 = float(input("Enter price of product 2: "))
quantity2 = int(input("Enter quantity of product 2: "))

price3 = float(input("Enter price of product 3: "))
quantity3 = int(input("Enter quantity of product 3: "))

if (budget < 0 or price1 <= 0 or price2 <= 0 or price3 <= 0
        or quantity1 <= 0 or quantity2 <= 0 or quantity3 <= 0):
    print("Invalid input")

else:
    cost1 = price1 * quantity1
    cost2 = price2 * quantity2
    cost3 = price3 * quantity3

    total = cost1 + cost2 + cost3

    print("Total Shopping Cost: Rs.", total)

    if total > budget:
        print("Status: Over Budget")
        print("Extra Amount Needed: Rs.", total - budget)
    else:
        print("Status: Within Budget")
        print("Remaining Budget: Rs.", budget - total)
```

output:

```
---------------------------------- RESTA
Enter shopping budget: 1000
Enter price of product 1: 500
Enter quantity of product 1: 1
Enter price of product 2: 400
Enter quantity of product 2: 1
Enter price of product 3: 300
Enter quantity of product 3: 1
Total Shopping Cost: Rs. 1200.0
Status: Over Budget
Extra Amount Needed: Rs. 200.0

================================== RESTA
Enter shopping budget: 2000
Enter price of product 1: 200
Enter quantity of product 1: 1
Enter price of product 2: 300
Enter quantity of product 2: 1
Enter price of product 3: 500
Enter quantity of product 3: 1
Total Shopping Cost: Rs. 1000.0
Status: Within Budget
Remaining Budget: Rs. 1000.0

================================== RESTA
Enter shopping budget: 2000
Enter price of product 1:
================================== RESTA
Enter shopping budget: -2000
Enter price of product 1: 200
Enter quantity of product 1: 1
Enter price of product 2: 300
Enter quantity of product 2: 1
Enter price of product 3: 900
Enter quantity of product 3: 1
Invalid input
```